

Stepwise Selection

STAT 223 – Applied Analytics

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April 24, 2025



Variable Selection

There are times when you want to know *which* variables in a set of p possible predictors.

If you *know* that a set of predictors are what you want to use to fit a model, you're set!

But often you have a set of variables that are *possibly* significant predictors, but you aren't sure which ones, if any, are.

This problem is often referred to as **variable selection** (a larger field), and more specifically **subset selection**.

Stepwise Regression

What we are going to discuss is an algorithm for choosing a model that attempts to select the optimal model based on some selection criterion, often p -values, R^2 or AIC/BIC.

Note that stepwise regression is not the perfect method for making a model. It has a habit of overfitting the model, or including noise variables/excluding important variables.

Don't let this be the be-all/end-all of your model fitting; instead, treat it as a step of your exploratory analysis. Try different methods of stepwise regression, combined with other methods and expert opinion to decide on the best set of predictors.

Stepwise Regression

There are three main methods for stepwise selection:

1. Forward Selection
2. Backward Selection
3. Stepwise Selection

Forward Selection

In **forward selection**, we start with an empty model, test adding each of the variables to the model individually, and add the one that adds the most.

So using AIC as our selection criterion, suppose we have a response variable: y ; and 5 possible explanatory variables: x_1 , x_2 , x_3 , x_4 and x_5 .

Forward Selection

Starting Model: $y = \beta_0$

Step 1:

Variable	AIC
None	28.348
x_1	-17.035
x_2	21.025
x_3	28.993
x_4	30.317
x_5	-48.994

Adding x_5 to the model reduces the AIC the most, so we add x_5 and move on to the next step.

Forward Selection

Starting Model: $y = \beta_0 + \beta_1 x_5$

Step 2:

Variable	AIC
None	-48.944
x_1	-217.959
x_2	-75.997
x_3	-46.951
x_4	-47.096

Adding x_1 to the model reduces the AIC the most, so we add x_1 and move on to the next step.

Forward Selection

Starting Model: $y = \beta_0 + \beta_1 x_5 + \beta_2 x_1$

Step 3:

Variable	AIC
None	-218.0
x_2	-7207.2
x_3	-216.0
x_4	-217.4

Adding x_2 to the model reduces the AIC the most, so we add x_2 and move on to the next step.

Forward Selection

Starting Model: $y = \beta_0 + \beta_1x_5 + \beta_2x_1 + \beta_3x_2$

Step 4:

Variable	AIC
None	-7202.2
x_3	-7207.8
x_4	-7211.4

Adding x_3 or x_4 doesn't significantly improve the AIC of the model compared to adding none, so we add neither and stop here.

Our final model is then:

$$y = \beta_0 + \beta_1 x_5 + \beta_2 x_1 + \beta_3 x_2$$

Backward Selection

In **backward selection**, we start with the full model, with all p predictors included, and test removing each of the variables from the model individually, and remove the one that adds the least.

So using AIC as our selection criterion, suppose we have a response variable: y ; and 5 possible explanatory variables: x_1, x_2, x_3, x_4 and x_5 .

Backward Selection

Starting Model: $y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_3 + \beta_4x_4 + \beta_5x_5$

Step 1:

Variable	AIC
None	-7407.6
x_1	-304.1
x_2	-7408.0
x_3	-7407.0
x_4	-7409.9
x_5	-7409.5

Removing x_4 from the model results in the model with the lowest AIC, so we remove that.

Backward Selection

Starting Model: $y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_3 + \beta_5x_5$

Step 2:

Variable	AIC
None	-7409.5
x_1	-304.1
x_2	-7410.0
x_3	-7409.0
x_5	-7411.5

Removing x_5 from the model results in the model with the lowest AIC, so we remove that.

Backward Selection

Starting Model: $y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_3$

Step 3:

Variable	AIC
None	-7411.5
x_1	-303.2
x_2	-7411.9
x_3	-7411.0

Removing x_2 from the model results in the model with the lowest AIC, so we remove that.

Backward Selection

Starting Model: $y = \beta_0 + \beta_1x_1 + \beta_3x_3$

Step 4:

Variable	AIC
None	-7411.9
x_1	-304.9
x_3	-7411.7

The current model has a lower AIC than the models removing x_1 or x_3 , so we finish the process here.

Backward Selection

Our final model is then:

$$y = \beta_0 + \beta_1 x_1 + \beta_3 x_3$$

Stepwise Selection

Both of these have their limitations, and one is that they only go in one direction.

There are cases where you can imagine where, for example, x_1 is a significant predictor for y on its own, but a combination of variables x_2 , x_3 and x_4 explain what x_1 explains better.

Forward selection would add x_1 first, and then possibly x_2 , x_3 and x_4 , and give a model with all four, leading to a model that is too large.

You can imagine a similar scenario with backwards selection.

Stepwise Selection

The solution for this is to try **stepwise selection**, which combines forward and backward selection.

You start with an empty model, add the variable that gives the best model, and then tests adding the other variables and removing any variables in the model, and then takes the best model resulting.

Keep doing that until the current model is better than any of the alternatives.